

CHRISTOPHER CAPIZZUTO

Software Engineer — Mobile & Full-Stack — ships products end to end: iOS and React Native clients; Next.js, Node, PostgreSQL and Firebase backends; and the tests and CI behind them. **Head of Chess** at a STEM nonprofit.

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TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Swift, Python, C++, SQL, HTML5/CSS3
 - **Mobile:** React Native, Expo SDK 54, SwiftUI, MapKit, CoreLocation, SwiftData, HealthKit, Xcode
 - **Frontend:** React 19, Next.js, Tailwind CSS, shadcn/ui, Radix UI, Vite, Webpack, Electron, Framer, Lottie/LottieLab
 - **Backend & Data:** Node.js, Express, FastAPI, REST, webhooks, AWS Lambda, Zod, PostgreSQL/PostGIS, MongoDB/Mongoose, Redis, Firebase (Auth, Firestore), Supabase, Azure SQL Database
 - **Cloud & DevOps:** AWS, Vercel, Cloudflare, Fly.io, Docker, Kubernetes (kind), GitHub Actions, Linux/Ubuntu
 - **Testing:** Playwright, Vitest, XCUITest, Swift Testing, Firebase Emulator Suite, axe-core, Postman
 - **Tools & AI:** Git/GitHub, Figma, Canva, Visual Studio | Claude Code, Anthropic SDK, Cursor, Codex, Gemini
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EXPERIENCE

Independent Mobile & Full-Stack Developer — Live Products · 2021 – Present

- Shipped and maintained 10+ live React Native and React applications, owning deployment, analytics, and production debugging end to end — including OAuth token lifecycles, webhook signature validation, and idempotent Stripe webhooks that hold state across retries.

Northeast STEM Starter Academy (NSSA) — Head of Chess · Mount Vernon, NY · Apr 2026 – Present

- Lead the chess program for a 501(c)(3) STEM education nonprofit, teaching three classes a week to elementary and middle school students.
- Designed and ran the three-week summer camp end to end — syllabus, daily lesson plans, and instruction differentiated across three skill-level classes against shared readiness benchmarks.
- Wrote the operating plan for a Saturday Chess Academy expansion for the founder and executive director: two skill-placed cohorts over a 32-Saturday term, a 9:1 student-to-adult ratio, readiness criteria before rated tournament play, outcome reporting, and a \$37,968 budget.

SparrowUp — Software Developer Intern (Stipended) · May 2025 – Oct 2025

- Turned motion-capture sensor output into coaching feedback — quantified seven biomechanical variables (strike speed, angle, rotation, knee bend, body position, plant-foot location, shot drive), scored against pro benchmarks across four stages of the movement.
 - Cut downstream video-analysis defects by documenting edge cases, test scenarios, and measurement inconsistencies pre-release.
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SELECTED PROJECTS

PrizeCheck — Pokémon TCG Practice Trainer · [\[Live\]](#) · [\[GitHub\]](#) — Next.js 16, React 19, TypeScript, Firebase, Zod, Docker, Kubernetes, Playwright, GitHub Actions

- Built a live trainer that simulates a real opening from a pasted deck list (shuffle, 8-card hand, 6 prizes) and scores recall on accuracy and speed, with lookups from a generated local index of 20,324 cards and an injection-hardened import parser.
- Shipped optional Google sign-in with saved decks and immutable deck revisions; Firestore rules enforce ownership, field bounds and write-once records, verified by emulator-run tests.
- Designed offline-tolerant sync: queued rounds persist in IndexedDB with stable session IDs so retries never duplicate, back off 1–60 seconds, and carry a history epoch so stale devices can't rewrite cleared history.
- Containerized the app (multi-stage, non-root, read-only Docker image) and ran two replicas on a local kind cluster with health probes and resource limits, testing pod replacement and rollback.

- Configured GitHub Actions to type-check and run unit, Firestore-rules and Playwright end-to-end tests (desktop, mobile, axe accessibility) against Firebase emulators, then build and smoke-test the container.

settle.ai — Room Planner with 3D Editor & Amazon Cart · [\[Live\]](#) — *Next.js 16, React 19, TypeScript, Three.js, Tailwind, Tesseract.js, PDF.js*

- Built a guided planner from a design handoff: users enter room measurements, drag doors and windows onto a live floor plan, pick a style and budget, get three valid layouts, and send the exact products to an Amazon cart.
- Wrote a deterministic layout engine — beam search over a 6-inch grid with ergonomic-access, circulation and collision checks, validated against 500 randomized rooms — and a raw Three.js editor with snapping, collision-safe moves and lazy-loaded 3D models.
- Extracts room dimensions from floor-plan images and PDFs entirely in the browser (Tesseract.js OCR, PDF.js), with versioned state migration, undo/redo, and Playwright visual-regression tests at desktop and 375px mobile.

FunkFit — Adaptive Nutrition for Lifters (iOS + Web) · [\[Live\]](#) · [\[GitHub\]](#) — *React Native 0.81, Expo SDK 54, TypeScript, HealthKit, Lottie, Fly.io*

- Shipped a cross-platform Expo app to iOS and web from one TypeScript codebase — Expo Router, haptics, Lottie animation, offline persistence, API credentials held off-device behind a Fly.io proxy, and Apple HealthKit sync behind a guarded native module so the same bundle still runs in Expo Go.
- Wrote a calorie engine driven by measured data rather than a formula: maintenance from wearable energy when present, an activity multiplier when not, clamped to safe floors, with goal dates fit to the real weigh-in trend and each target recording its source.
- Traced two blank-screen production failures on web — a style array React DOM rejects, injected by a global Text patch, and fonts 404ing into an SPA rewrite — and made font loading fall back to system fonts.

Pidge — NYC Parking Intelligence (Native iOS + Backend) · [\[GitHub\]](#) — *SwiftUI, MapKit, CoreLocation, SwiftData, Express, TypeScript, PostgreSQL/PostGIS, MapLibre*

- Built native SwiftUI + MapKit and React/MapLibre clients that turn NYC curb rules into a color-coded parking map, backed by an Express/PostGIS API — Zod validation, seven staged migrations, viewport-indexed spatial queries, and scheduled ingestion of NYC Open Data signs, meters and curb geometry.
- Designed it to fail closed — unknown and no-parking curbs are never recommended — and enforced that in tests: XCUITest and Swift Testing on iOS, Vitest on the backend, and a 9-case Playwright suite proving the map never fabricates an option.

backstop.ai — Live Catcher Analytics + AI Analyst · [\[Live\]](#) · [\[GitHub\]](#) — *Next.js, TypeScript, Python, FastAPI, PostgreSQL, Claude API*

- Grades MLB catchers' pitch-calling from public Statcast data — Python models score each call against realistic pitcher-specific alternatives plus framing, blocking, pop time and arm strength, served via FastAPI.
- Added a live AI analyst: Claude streams token by token from a Next.js route handler, grounded strictly in the on-page zone report (no invented stats) and cached per pitch count so polling never re-bills.

Also built: **Dragapultist** ([live](#)) — MongoDB aggregation pipelines for win rates and matchup stats from Pokémon TCG game logs, plus an Electron build · **CareCation** ([live](#)) — medical travel booking with Amadeus flight/hotel search, Stripe payments and Supabase · **Hudson Chess** (hudsonchess.com) — Vite + React chess academy site and blog

EDUCATION & CERTIFICATIONS

Georgia Institute of Technology — M.S. Computer Science, AI specialization (incoming)

Codesmith — Full-Stack Software Engineering

Baruch College (CUNY) — B.B.A., Digital Marketing

AWS Certified DevOps Engineer – Professional — in progress

Claude Certified Architect – Professional (Anthropic) — in progress